

circuit is built around microcontroller ATmega8 (IC1), temperature sensor LM35 (IC2), thermoelectric module TEC1-12706 (connected at CON4), desktop computer's SMPS (connected at CON1), DC fan (connected at CON3), 7-segment displays (DIS1 and DIS2) and relay (RL1).

The complete circuit is powered by desktop computer's SMPS. 12V and 5V outputs from the SMPS are connected at CON1. 5V supply is used to power microcontroller IC1 and rest of the circuit, while 12V is used for the DC fan and thermoelectric module.

Thermoelectric module. Actual cooling is done by thermoelectric module TEC1-12706. It works on the principle of Seebeck effect: when current is passed through two dissimilar metal junctions, one junction gets heated while the other junction cools down.

The module is 40×40×3.6mm in size. Although it operates over 4V-16V, recommended operating voltage is 12V. Depending on temperature of hot side, TEC1-12706 is capable of transferring 50-60W of heat. Outer construction of this module is ceramic and the metal junctions are inside along two surfaces.



Fig. 3: Thermoelectric module

As shown

in Fig. 3, the side on which model number is printed is the hot side and the other one is cold side.

Polarity of voltage is very important; red wire should be connected to 12V and black to ground. If the polarity is reversed, hot and cold sides get reversed as well. Heat produced on the hot side must be dissipated, otherwise the module will heat up beyond limit. We can make use of any heat-sink such as the one used for CPU in a computer. Temperature of hot side has direct effect on how much heat the module can suck from the cold side. Here we have mounted the DC fan over the heat-sink to dissipate heat more efficiently.

Microcontroller. ATmega8 is a low-power CMOS 8-bit microcontroller based on the AVR RISC architecture with throughputs approaching 1MIPS per megahertz. ATmega8 comes with the 8 kilobytes of in-system programmable Flash, 512 bytes of EEPROM, 1 kilobyte of SRAM, 23 general-purpose I/O lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a byte oriented two-wire serial interface, a 6-channel ADC (eight channels in TQFP and QFN/MLF packages) with 10-bit accuracy, a programmable watchdog timer with internal oscillator and an SPI serial port.

Microcontroller IC1 runs at a clock frequency of 8MHz using an internal

oscillator. The thermoelectric module sucks heat from the water inside the refrigerator box and keeps cooling it down. The temperature of water is sensed through temperature sensor IC2 which is interfaced to the microcontroller's ADC pin 24. When the temperature reaches 9°C or above, relay RL1 gets energised and switches on the thermoelectric module. When the temperature reaches 5°C or below, the relay is de-energised and the thermoelectric module is switched off.

Port pins PD0 through PD7 of IC1 are connected to pins g, f, a, b, dp, c, d and e of 7-segment displays, DIS1 and DIS2. Port pins PB1 and PB2 of IC1 drive the base of transistors T1 and T2 for switching on 7-segment displays, DIS1 and DIS2, respectively. The 7-segment displays show the temperature sensed by IC2.

Software

The software program is written in 'C' programming language and compiled using AVR Studio. The program is burnt in the microcontroller using a suitable programmer.

Construction and testing

An actual-size, single-side PCB for the thermoelectric refrigerator circuit is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on the PCB to minimise any assembly errors. Use IC base for microcontroller

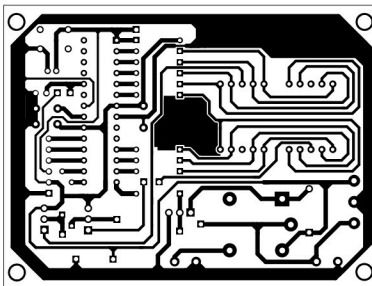


Fig. 4: An actual-size, single-side PCB for the thermoelectric refrigerator

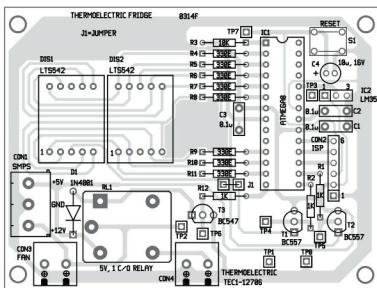


Fig. 5: Component layout for the PCB